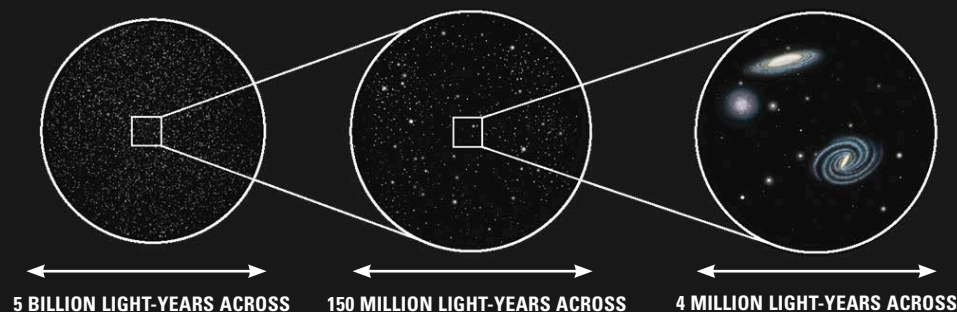


FILAMENTS AND VOIDS

At the largest scales measured, the universe reveals a clear overall structure. Galaxy superclusters join to form stringlike “filaments” or flat “sheets” around the edges of enormous and apparently empty regions known as “voids.” While galaxy structure up to the level of clusters can be explained by the action of gravity since the Big Bang, the present age of the universe (13.7 billion years) is not nearly long enough for gravity alone to have organized the universe on the scale of filaments and voids. This indicates that large-scale cosmic structures are, in fact, expanded “echoes” of features from the earliest times. The first filaments to be discovered were galactic “walls” identified in the 1980s, and since then it has become clear that filaments contain not only luminous galaxies but also enormous clouds of hydrogen known as Lyman Alpha blobs. The first void, meanwhile, was discovered during a galaxy survey in 1978. Typically, voids are empty of both normal and dark matter, although some voids have been found to contain a few galaxies.



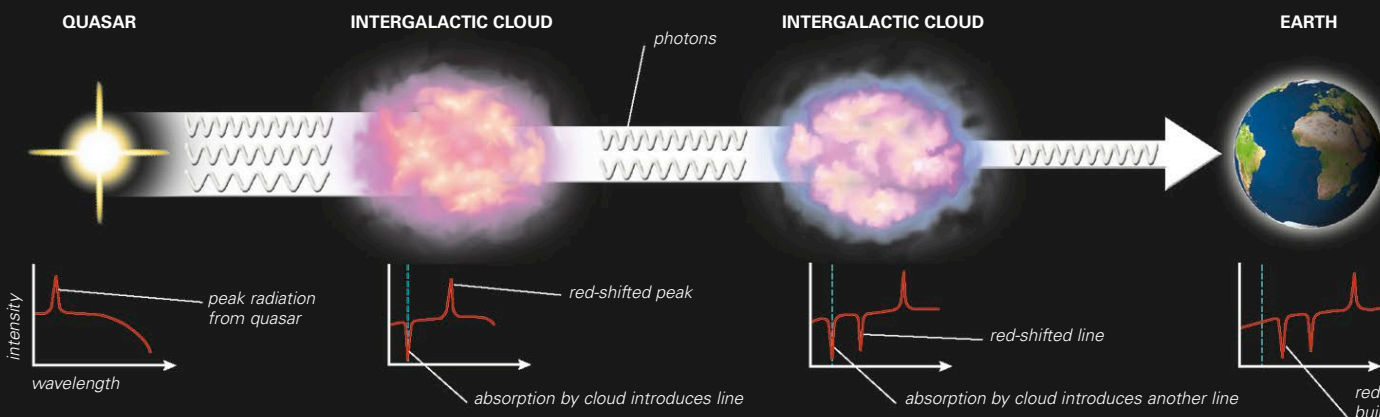
THE COSMOLOGICAL PRINCIPLE

This principle is the assumption that, at the largest scales, the universe is essentially uniform in all its properties and in all directions, even though it is clearly not so at smaller scales. The principle seems to be borne out in practice—for example, when comparing the distribution of galaxies, as shown above.

unmapped sections are areas blocked from the telescope's view by the Milky Way

BETWEEN THE SUPERCLUSTERS

Studying luminous galaxies alone can give a deceptive view of the universe—not all normal matter produces detectable radiation, and dark matter neither produces radiation nor interacts with it. However, by analyzing light from distant quasars (see p.320), astronomers can measure the effects of intervening, but otherwise invisible, hydrogen clouds. As light from the quasars passes through such clouds, the hydrogen “imprints” it with absorption lines that form a pattern called a Lyman Alpha forest. The wavelengths of these absorption lines reveal the red shifts of the clouds and therefore their distance from Earth, allowing their distribution to be mapped. In addition, analyzing localized movements among galaxy clusters allows astronomers to map the distribution of dark matter. Both of these methods seem to confirm that the voids between superclusters are empty and that most normal and dark matter is concentrated around the visible galaxy filaments.



LYMAN-ALPHA FOREST

Light from a distant quasar passes through a series of hydrogen clouds on its way to Earth. Each superimposes a Lyman Alpha absorption line onto the quasar's spectrum, but red shifts mean the lines do not overlap. The result is a series of red-shifted lines called a Lyman-Alpha forest.

GALAXY MAP FROM THE SLOAN SURVEY

Begun in 2000, the Sloan Digital Sky Survey is a major red-shift survey that has so far mapped more than a million objects, out to 2 billion light-years from Earth. On this map, each dot represents a galaxy, and the galaxies are plotted at distances from the center proportional to their distance from Earth.

each dot is a separate galaxy, color-coded according to the average age of its stars: red dots contain older stars; blue and green are younger stars

the edge of the survey map is about 2 billion light-years away from the Milky Way

dark regions in the mapped area are huge voids in space

filamentary structures are strings of galaxy clusters that are only partially mapped